

Medicare

use medicaredb;

Tables:

show tables;

```
33455
33456
33457 • show tables;
33458
33459
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Tables_in_medicaredb			
admissions			
appointments			
billing			
departments			
doctors			
employees			
hospitals			
insurance			
laboratory			
medicines			
patients			
payments			
pharmacy			
rooms			
treatments			

Data profiling:

Departments

SELECT * from Departments where head_doctor_id is null;

```
33459
33460 /* Departments data profiling */
33461
33462 • SELECT * from Departments where head_doctor_id is null;
33463
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
department_id	department_name	hospital_id	floor_number	head_doctor_id
D002	Emergency Medicine	H002	4	NULL
D005	Neurology	H004	5	NULL
D006	Ophthalmology	H005	2	NULL
D022	ENT	H017	3	NULL
D026	General Medicine	H020	6	NULL
D027	ENT	H020	6	NULL
D030	Endocrinology	H022	3	NULL
D032	Nephrology	H024	3	NULL
D035	Gynecology	H026	1	NULL
D036	Dermatology	H027	4	NULL
D038	Nephrology	H028	5	NULL
NULL	NULL	NULL	NULL	NULL

Doctors

select distinct gender
from doctors;

```
33465
33466      /* Doctors data profiling */
33467
33468 •   select distinct gender
33469      from doctors;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	gender			
▶	Female			
	Male			

select gender,count(*)
from doctors
group by gender;

```
33468
33469 •   select gender,count(*)
33470      from doctors
33471      group by gender;
33472
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	gender	count(*)			
▶	Female	231			
	Male	219			

```
select count(distinct gender) as uni_gc
from doctors;
```

```
33472
33473 • select count(distinct gender) as uni_gd
33474 from doctors;
33475
33476
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	uni_gc
▶	2

```
SELECT * FROM Doctors where department_id IS NULL;
```

```
33476 |
33477 |
33478 | • SELECT * FROM Doctors where department_id IS NULL;
33479 |
33480 |
```

[illegible]

```
SELECT * FROM Doctors where email IS NULL;
```

```
13479
13480
13481 • SELECT * FROM Doctors where email IS NULL;
13482
13483
```

Result Grid												
Filter Rows: Edit: Export/Import: Wrap Cell Contents:												
doctor_id	first_name	last_name	gender	specialization	department_id	hospital_id	qualification	experience_years	consultation_fee	phone_number	email	joining_date
DR0006	Bhavani	Reddappa	Female	Nephrology	D021	H016	MBBS, FRCS	2	1439.53	9464170805		1999-10-22
DR0075	Ishwarya	Rao	Female	Dermatology	D036	H027	MBBS, MCh	3	644.51	9064039106		2001-02-04

SELECT * FROM Doctors where email NOT REGEXP
'^[A-Za-z0-9_%.~]+@[A-Za-z0-9_~]+\.[A-Za-z]{2,}\$';

```
33483
33484
33485 • SELECT * FROM Doctors where email NOT REGEXP '^[A-Za-z0-9_%.~]+@[A-Za-z0-9_~]+\.[A-Za-z]{2,}$';
33486
33487
```

doctor_id	first_name	last_name	gender	specialization	department_id	hospital_id	qualification	experience_years	consultation_fee	phone_number	email	joining_date
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Patients

SELECT * from Patients where first_name <> ltrim(first_name);

```
33534
33535 /* Patients data profiling */
33536
33537 • SELECT * from Patients where first_name <> ltrim(first_name);
33538
```

patient_id	first_name	last_name	gender	date_of_birth	age	city	state	phone_number	email	blood_group	registration_date
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

SELECT gender, count(*)
FROM Patients
group by gender;

```
33539
33540 • SELECT gender, count(*)
33541 FROM Patients
33542 group by gender;
33543
```

gender	count(*)
Male	1003
Female	997

```
33543
33544 • SELECT * FROM Patients where email is null;
33545
33546
33547
```

```
SELECT * from Patients where email NOT like '%@%.%';
```

Admission

SELECT * from Admissions where department_id is null;

```
33600  /* Admission Data Profiling */
```

```
33601
```

```
33602 • SELECT * from Admissions where department_id is null;
```

```
33603
```

Result Grid										
Filter Rows: Edit: Export/Import: Wrap Cell Content:										
admission_id	patient_id	hospital_id	department_id	admitting_doctor_id	room_id	admission_date	discharge_date	admission_type	admission_status	
AD00121	PT000367	H018	NULL	DR0152	RM00499	2024-05-17	NULL	Emergency	Under Observation	
AD00151	PT001680	H029	NULL	DR0054	RM00747	2024-01-23	2024-01-31	Planned	Discharged	
AD00164	PT000992	H028	NULL	DR0301	RM00724	2025-01-17	NULL	Emergency	Discharged	
AD00172	PT000592	H013	NULL	DR0296	RM00364	2025-02-13	NULL	Referral	Admitted	
AD00195	PT000495	H001	NULL	DR0403	RM00001	2025-08-06	NULL	Planned	Admitted	
AD00213	PT001012	H023	NULL	DR0307	RM00590	2024-05-06	2024-05-12	Referral	Discharged	
AD00236	PT000021	H023	NULL	DR0395	RM00599	2024-05-18	NULL	Emergency	Admitted	
AD00260	PT000983	H012	NULL	DR0331	RM00344	2025-08-09	NULL	Emergency	Under Observation	
AD00278	PT000116	H015	NULL	DR0001	RM00442	2025-06-21	2025-06-22	Planned	Discharged	
AD00303	PT001197	H023	NULL	DR0395	RM00598	2024-05-29	2024-05-30	Emergency	Discharged	
AD00355	PT000414	H029	NULL	DR0280	RM00753	2024-12-31	NULL	Planned	Admitted	
AD00358	PT000785	H015	NULL	DR0138	RM00451	2024-12-20	NULL	Emergency	Admitted	
AD00384	PT000393	H015	NULL	DR0052	RM00439	2024-09-01	2024-09-09	Emergency	Discharged	
AD00416	PT000496	H001	NULL	DR0403	RM00003	2024-07-23	NULL	Planned	Under Observation	
AD00531	PT000196	H019	NULL	DR0112	RM00543	2025-01-25	2025-02-06	Emergency	Discharged	
AD00541	PT001714	H003	NULL	DR0339	RM00034	2025-06-09	2025-06-10	Planned	Discharged	
AD00613	PT001582	H016	NULL	DR0215	RM00465	2024-02-10	NULL	Emergency	Under Observation	
AD00643	PT001257	H008	NULL	DR0426	RM00197	2025-09-20	2025-09-28	Emergency	Discharged	
AD00680	PT000750	H017	NULL	DR0045	RM00472	2025-12-16	2025-12-30	Emergency	Discharged	
AD00717	PT000742	H023	NULL	DR0274	RM00596	2025-08-13	NULL	Planned	Under Observation	
AD00734	PT001038	H024	NULL	DR0348	RM00609	2025-04-14	NULL	Planned	Admitted	
AD00753	PT001847	H019	NULL	DR0273	RM00550	2025-03-19	2025-04-02	Planned	Discharged	
AD00784	PT001800	H030	NULL	DR0255	RM00766	2025-11-21	2025-11-24	Referral	Discharged	
AD00788	PT000033	H008	NULL	DR0123	RM00733	2025-05-08	NULL	Emergency	Under Observation	

SELECT * from Admissions where discharge_date is null;

```
33606 • SELECT * from Admissions where discharge_date is null;
```

```
33607
```

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33608
```

Result Grid										
Filter Rows: Edit: Export/Import: Wrap Cell Content:										
admission_id	patient_id	hospital_id	department_id	admitting_doctor_id	room_id	admission_date	discharge_date	admission_type	admission_status	
AD00001	PT001018	H015	D019	DR0043	RM00449	2024-05-27	NULL	Referral	Admitted	
AD00003	PT001895	H008	D010	DR0409	RM00195	2024-08-15	NULL	Planned	Under Observation	
AD00004	PT000925	H009	D012	DR0407	RM00213	2025-06-27	NULL	Referral	Admitted	
AD00009	PT001240	H011	D014	DR0038	RM00307	2024-03-06	NULL	Planned	Under Observation	
AD00016	PT000353	H007	D009	DR0027	RM00167	2024-12-13	NULL	Referral	Admitted	
AD00017	PT000698	H015	D019	DR0163	RM00447	2024-03-22	NULL	Emergency	Admitted	
AD00018	PT000874	H014	D018	DR0288	RM00405	2024-03-30	NULL	Emergency	Admitted	
AD00025	PT001131	H009	D012	DR0261	RM00227	2025-05-17	NULL	Emergency	Admitted	
AD00026	PT000654	H026	D035	DR0040	RM00673	2024-10-17	NULL	Referral	Under Observation	
AD00028	PT000446	H013	D017	DR0007	RM00374	2025-05-22	NULL	Emergency	Admitted	
AD00029	PT001187	H014	D018	DR0355	RM00424	2024-01-13	NULL	Planned	Discharged	
AD00030	PT001316	H005	D006	DR0277	RM00095	2024-09-03	NULL	Planned	Under Observation	
AD00047	PT001227	H015	D019	DR0165	RM00444	2024-11-28	NULL	Referral	Admitted	
AD00049	PT000467	H004	D005	DR0008	RM00076	2024-06-02	NULL	Referral	Admitted	
AD00052	PT000191	H008	D010	DR0275	RM00198	2024-03-27	NULL	Emergency	Under Observation	
AD00053	PT001808	H005	D006	DR0277	RM00098	2024-05-05	NULL	Emergency	Discharged	
AD00055	PT000413	H027	D036	DR0397	RM00700	2024-10-05	NULL	Emergency	Under Observation	
AD00057	PT000219	H017	D022	DR0133	RM00479	2024-04-27	NULL	Emergency	Discharged	
AD00071	PT000458	H003	D003	DR0328	RM00059	2025-05-31	NULL	Planned	Admitted	
AD00073	PT001961	H024	D032	DR0324	RM00615	2024-12-01	NULL	Referral	Under Observation	
AD00074	PT000348	H008	D010	DR0275	RM00204	2025-10-27	NULL	Emergency	Discharged	
AD00078	PT001499	H015	D020	DR0258	RM00443	2024-06-23	NULL	Emergency	Admitted	
AD00079	PT001994	H012	D015	DR0347	RM00330	2025-07-18	NULL	Emergency	Admitted	
AD00088	PT001215	H003	D004	DR0048	RM00045	2024-10-07	NULL	Emergency	Admitted	

Treatments

SELECT * FROM Treatments where Admission_id is null;

```
33609
33610 /* Treatments Data Profiling */
33611
33612 • SELECT * FROM Treatments where Admission_id is null;
33613
```

admission_id	patient_id	hospital_id	department_id	admitting_doctor_id	room_id	admission_date	discharge_date	admission_type	admission_status
AD00001	PT001018	H015	D019	DR0043	RM00449	2024-05-27	NULL	Referral	Admitted
AD00003	PT001895	H008	D010	DR0409	RM00195	2024-08-15	NULL	Planned	Under Observation
AD00004	PT000925	H009	D012	DR0407	RM00213	2025-06-27	NULL	Referral	Admitted
AD00009	PT001240	H011	D014	DR0038	RM00307	2024-03-06	NULL	Planned	Under Observation
AD00016	PT000353	H007	D009	DR0027	RM00167	2024-12-13	NULL	Referral	Admitted
AD00017	PT000698	H015	D019	DR0163	RM00447	2024-03-22	NULL	Emergency	Admitted
AD00018	PT000874	H014	D018	DR0288	RM00405	2024-03-30	NULL	Emergency	Admitted
AD00025	PT001131	H009	D012	DR0261	RM00227	2025-05-17	NULL	Emergency	Admitted
AD00026	PT000654	H026	D035	DR0040	RM00673	2024-10-17	NULL	Referral	Under Observation
AD00028	PT000446	H013	D017	DR0007	RM00374	2025-05-22	NULL	Emergency	Admitted
AD00029	PT001187	H014	D018	DR0355	RM00424	2024-01-13	NULL	Planned	Discharged
AD00030	PT001316	H005	D006	DR0277	RM00095	2024-09-03	NULL	Planned	Under Observation
AD00047	PT001227	H015	D019	DR0165	RM00444	2024-11-28	NULL	Referral	Admitted
AD00049	PT000467	H004	D005	DR0008	RM00076	2024-06-02	NULL	Referral	Admitted
AD00052	PT000191	H008	D010	DR0275	RM00198	2024-03-27	NULL	Emergency	Under Observation
AD00053	PT001808	H005	D006	DR0277	RM00098	2024-05-05	NULL	Emergency	Discharged
AD00055	PT000413	H027	D036	DR0397	RM00700	2024-10-05	NULL	Emergency	Under Observation
AD00057	PT000219	H017	D022	DR0133	RM00479	2024-04-27	NULL	Emergency	Discharged
AD00071	PT000458	H003	D003	DR0328	RM00059	2025-05-31	NULL	Planned	Admitted
AD00073	PT001961	H024	D032	DR0324	RM00615	2024-12-01	NULL	Referral	Under Observation
AD00074	PT000348	H008	D010	DR0275	RM00204	2025-10-27	NULL	Emergency	Discharged
AD00078	PT001499	H015	D020	DR0258	RM00443	2024-06-23	NULL	Emergency	Admitted
AD00079	PT001994	H012	D015	DR0347	RM00330	2025-07-18	NULL	Emergency	Admitted
AD00088	PT001215	H003	D004	DR0048	RM00045	2024-10-07	NULL	Emergency	Admitted

Insurance

SELECT * FROM Insurance where insurance_provider is null;

```
33613
33614 /* Insurance Data Profiling */
33615
33616 • SELECT * FROM Insurance where insurance_provider is null;
33617
```

insurance_id	patient_id	insurance_provider	policy_number	coverage_amount	policy_start_date	policy_end_date	claim_status
IN00062	PT000094	NULL	POL721378	500000	2021-02-28	2022-02-28	Pending
IN00093	PT001338	NULL	POL884783	200000	2023-09-26	2024-09-26	Settled
IN00124	PT001172	NULL	POL382730	500000	2020-04-08	2021-04-08	Approved
IN00137	PT000975	NULL	POL604943	750000	2021-12-10	2022-12-10	Rejected
IN00145	PT000555	NULL	POL644807	500000	2020-05-31	2021-05-31	Pending
IN00149	PT001790	NULL	POL394616	300000	2021-09-30	2022-09-30	Approved
IN00157	PT001971	NULL	POL755271	200000	2022-05-04	2023-05-04	Approved
IN00161	PT000017	NULL	POL389665	200000	2022-03-11	2023-03-11	Pending
IN00165	PT001653	NULL	POL143752	1000000	2021-06-21	2022-06-21	Rejected
IN00180	PT001148	NULL	POL239376	300000	2022-05-14	2023-05-14	Approved
IN00196	PT000379	NULL	POL235385	750000	2024-08-14	2025-08-14	Approved
IN00202	PT001729	NULL	POL251440	500000	2023-04-13	2024-04-13	Pending
IN00208	PT000598	NULL	POL283296	200000	2020-09-15	2021-09-15	Settled
IN00230	PT001941	NULL	POL691359	500000	2024-08-16	2025-08-16	Settled
IN00258	PT001394	NULL	POL603965	300000	2023-11-04	2024-11-04	Approved
IN00268	PT000724	NULL	POL516472	500000	2021-07-19	2022-07-19	Pending
IN00276	PT000522	NULL	POL921142	1000000	2023-09-30	2024-09-30	Rejected
IN00296	PT000448	NULL	POL348839	300000	2020-08-12	2021-08-12	Approved
IN00298	PT000786	NULL	POL637583	200000	2024-06-25	2025-06-25	Rejected
IN00314	PT000097	NULL	POL685942	750000	2024-11-03	2025-11-03	Approved
IN00342	PT000875	NULL	POL122864	200000	2023-05-18	2024-05-18	Settled
IN00354	PT000705	NULL	POL481684	300000	2023-03-24	2024-03-24	Pending
IN00384	PT000761	NULL	POL824033	300000	2021-10-09	2022-10-09	Settled
IN00389	PT001195	NULL	POL489668	300000	2021-10-07	2022-10-07	Pending

Employees

SELECT gender, count(*)
from Employee
group by gender;

```
33618  /* Employees Data Profiling */
33619
33620  •  SELECT gender, count(*)
33621  from Employee
33622  group by gender;
```

	insurance_id	patient_id	insurance_provider	policy_number	coverage_amount	policy_start_date	policy_end_date	claim_status
▶	IN00062	PT000094	NULL	POL721378	500000	2021-02-28	2022-02-28	Pending
	IN00093	PT001338	NULL	POL884783	200000	2023-09-26	2024-09-26	Settled
	IN00124	PT001172	NULL	POL382730	500000	2020-04-08	2021-04-08	Approved
	IN00137	PT000975	NULL	POL604943	750000	2021-12-10	2022-12-10	Rejected
	IN00145	PT000555	NULL	POL644807	500000	2020-05-31	2021-05-31	Pending
	IN00149	PT001790	NULL	POL394616	300000	2021-09-30	2022-09-30	Approved
	IN00157	PT001971	NULL	POL755271	200000	2022-05-04	2023-05-04	Approved
	IN00161	PT000017	NULL	POL389665	200000	2022-03-11	2023-03-11	Pending
	IN00165	PT001653	NULL	POL143752	1000000	2021-06-21	2022-06-21	Rejected
	IN00180	PT001148	NULL	POL239376	300000	2022-05-14	2023-05-14	Approved
	IN00196	PT000379	NULL	POL235385	750000	2024-08-14	2025-08-14	Approved
	IN00202	PT001729	NULL	POL251440	500000	2023-04-13	2024-04-13	Pending
	IN00208	PT000598	NULL	POL283296	200000	2020-09-15	2021-09-15	Settled
	IN00230	PT001941	NULL	POL691359	500000	2024-08-16	2025-08-16	Settled
	IN00258	PT001394	NULL	POL603965	300000	2023-11-04	2024-11-04	Approved
	IN00268	PT000724	NULL	POL516472	500000	2021-07-19	2022-07-19	Pending
	IN00276	PT000522	NULL	POL921142	1000000	2023-09-30	2024-09-30	Rejected
	IN00296	PT000448	NULL	POL348839	300000	2020-08-12	2021-08-12	Approved
	IN00298	PT000786	NULL	POL637583	200000	2024-06-25	2025-06-25	Rejected
	IN00314	PT000097	NULL	POL685942	750000	2024-11-03	2025-11-03	Approved
	IN00342	PT000875	NULL	POL122864	200000	2023-05-18	2024-05-18	Settled
	IN00354	PT000705	NULL	POL481684	300000	2023-03-24	2024-03-24	Pending
	IN00384	PT000761	NULL	POL824033	300000	2021-10-09	2022-10-09	Settled

SELECT * FROM Employees where department_id is null;

```
33626  •  SELECT * FROM Employees where department_id is null;
33627
33628  /* Employees Data Profiling */
```

	employee_id	first_name	last_name	gender	hospital_id	department_id	designation	employment_type	salary	joining_date	phone_number	email
▶	EMP00038	Lokesh	Chetty	Male	H005	NULL	Lab Technician	Part-Time	111867.08	2012-03-25	9196495170	lokesh.chet
	EMP00046	Mohan	Iyengar	Male	H005	NULL	Housekeeping Staff	Part-Time	53575.28	2016-03-25	9847508961	mohan.iyer
	EMP00064	Vani	Subramaniam	Female	H018	NULL	IT Support	Part-Time	66944.35	2011-10-12	9434674071	vani.subrar
	EMP00072	Arun	Iyengar	Male	H021	NULL	Accountant	Full-Time	84503.72	2009-07-18	9778171743	arun.iyeng
	EMP00082	Prakash	Reddy	Male	H017	NULL	Security Staff	Part-Time	56597.70	2012-11-14	9705924686	prakash.re
	EMP00083	Krishnan	Venkatesh	Male	H015	NULL	Housekeeping Staff	Full-Time	33583.47	2007-06-05	9148244178	krishnan.ve
	EMP00091	Mahesh	Iyer	Male	H029	NULL	Administrator	Full-Time	105182.09	2017-06-05	9294692417	mahesh.iye
	EMP00098	Abhinav	Mudaliar	Male	H001	NULL	IT Support	Part-Time	49206.90	2009-09-17	9232885793	abhinav.m
	EMP00119	Bhavya	Hegde	Female	H002	NULL	Radiology Technician	Part-Time	73903.26	2023-11-01	9194056303	bhavya.he
	EMP00124	Yamuna	Hegde	Female	H024	NULL	Housekeeping Staff	Full-Time	91593.47	2018-06-06	9116104934	yamuna.he
	EMP00125	Sathish	Menon	Male	H015	NULL	Accountant	Full-Time	59219.74	2017-03-30	9651990304	sathish.me
	EMP00138	Ramesh	Chetty	Male	H017	NULL	Radiology Technician	Full-Time	27346.62	2012-11-11	9387673209	ramesh.ch
	EMP00147	Sandhya	Subramaniam	Female	H027	NULL	Pharmacist	Full-Time	99668.27	2014-09-19	9326884524	sandhya.s
	EMP00150	Prakash	Sundaram	Male	H002	NULL	Staff Nurse	Full-Time	22420.30	2005-05-05	9860189533	prakash.su
	EMP00151	Sandeep	Mudaliar	Male	H025	NULL	Receptionist	Full-Time	84938.34	2006-04-26	9716921226	sandeep.m
	EMP00158	Usha	Gopalakrishnan	Female	H030	NULL	IT Support	Full-Time	57098.67	2014-05-18	9252707980	usha.gopal
	EMP00160	Krishnan	Gopalakrishnan	Male	H030	NULL	Accountant	Full-Time	61149.59	2007-12-05	9005475611	krishnan.g
	EMP00195	Ravi	Paricker	Male	H016	NULL	Pharmacist	Full-Time	89649.53	2006-12-08	9618563374	ravi.panick
	EMP00201	Keerthana	Shetty	Female	H017	NULL	Radiology Technician	Full-Time	47020.56	2015-07-10	9014958457	keerthana.
	EMP00213	Bhavya	Menon	Female	H005	NULL	Radiology Technician	Full-Time	106064.17	2013-03-05	9675443496	bhavya.me
	EMP00237	Prakash	Mudaliar	Male	H028	NULL	Ward Boy	Contract	47103.10	2023-05-06	9159153796	prakash.m
	EMP00241	Latha	Murthv	Male	H022	NULL	Staff Nurse	Full-Time	91802.47	2025-07-19	9603432718	latha.murth

Billing

SELECT * FROM Billing where admission_id is null;

```
33642
33643 /* Billing Data Profilling */
33644
33645 • SELECT * FROM Billing where admission_id is null;
33646
```

bill_id	patient_id	admission_id	appointment_id	bill_date	room_charges	doctor_charges	medicine_charges	lab_charges	other_charges	total_amount	bill_status
BL000001	PT000782	NULL	AP004939	2025-09-06	0.00	2718.07	625.68	222.12	1025.67	6428.16	Paid
BL000003	PT001972	NULL	AP000894	2024-01-31	0.00	3466.81	1920.19	2735.17	558.26	8680.43	Paid
BL000005	PT000786	NULL	AP001968	2025-01-04	0.00	316.45	892.32	3711.74	1296.88	6217.39	Partially P
BL000010	PT001173	NULL	AP004280	2024-11-25	0.00	451.86	2649.58	4404.25	152.70	10721.75	Partially P
BL000011	PT001773	NULL	AP001897	2025-12-09	0.00	2060.40	536.24	869.56	670.98	4137.18	Paid
BL000016	PT001865	NULL	AP004526	2024-05-01	0.00	2228.31	1469.27	2769.48	311.36	12201.16	Paid
BL000017	PT001853	NULL	AP003608	2025-10-18	0.00	3931.50	3007.72	4536.36	113.26	11588.84	Unpaid
BL000018	PT000294	NULL	AP003946	2025-02-26	0.00	4322.11	3567.77	1490.75	656.87	18067.50	Partially P
BL000020	PT000210	NULL	AP003237	2025-01-10	0.00	2199.74	3044.99	3293.89	1341.59	9880.21	Paid
BL000021	PT001939	NULL	AP000446	2024-06-22	0.00	3628.80	2375.21	837.85	336.07	7177.93	Unpaid
BL000022	PT000427	NULL	AP004057	2024-12-14	0.00	593.37	189.98	2639.35	1107.92	4530.62	Paid
BL000026	PT001160	NULL	AP001700	2025-04-19	0.00	3569.69	810.85	2415.49	1433.02	8229.05	Partially P
BL000027	PT001733	NULL	AP003658	2025-02-05	0.00	3867.31	1555.17	3506.04	1085.48	10014.00	Paid
BL000028	PT001313	NULL	AP000002	2024-02-21	0.00	3975.39	1262.52	5965.96	138.55	11342.42	Paid
BL000030	PT001795	NULL	AP004471	2024-03-27	0.00	4872.27	2095.39	2927.37	1147.33	11042.36	Partially P
BL000031	PT001732	NULL	AP001839	2025-11-11	0.00	1809.67	2976.65	5119.31	954.10	10859.73	Paid
BL000032	PT000692	NULL	AP004096	2025-06-12	0.00	745.53	1385.51	5060.35	827.34	11226.22	Unpaid
BL000034	PT000212	NULL	AP004868	2024-06-14	0.00	890.27	3262.38	4344.53	376.49	12423.14	Paid
BL000035	PT000363	NULL	AP004822	2025-12-11	0.00	3118.62	1425.40	2378.94	1134.30	11280.16	Paid
BL000036	PT000998	NULL	AP002630	2025-08-15	0.00	2069.09	3848.25	3441.17	268.36	13477.62	Paid
BL000037	PT000082	NULL	AP000911	2025-08-30	0.00	4703.21	1260.77	2511.21	1262.85	17528.47	Paid
BL000038	PT001015	NULL	AP002349	2024-06-06	0.00	2906.61	432.81	2687.95	905.12	6932.49	Paid
BL000039	PT001257	NULL	AP004546	2024-09-15	0.00	1588.83	1700.50	707.78	477.88	6473.78	Paid

SELECT * FROM Billing where appointment_id is null;

```
3648 • SELECT * FROM Billing where appointment_id is null;
3649
3650
```

bill_id	patient_id	admission_id	appointment_id	bill_date	room_charges	doctor_charges	medicine_charges	lab_charges	other_charges	total_amount	bill_status
BL000002	PT000636	AD01321	NULL	2024-03-09	14472.59	4266.95	7.60	858.25	246.34	50026.36	Unpaid
BL000004	PT000714	AD01867	NULL	2025-01-25	14493.34	2896.80	3508.29	2834.99	771.69	24505.11	Partially P
BL000006	PT001628	AD02078	NULL	2025-10-24	7635.34	2179.54	2476.28	1358.89	37.68	13687.73	Partially P
BL000007	PT001126	AD02315	NULL	2025-05-10	5846.27	4203.28	2797.47	1279.13	1016.06	21199.09	Unpaid
BL000008	PT000988	AD02160	NULL	2025-11-11	7486.42	4935.34	841.19	3141.67	935.24	24275.80	Paid
BL000009	PT000093	AD02017	NULL	2024-08-26	9579.41	771.57	3124.84	380.76	805.52	14662.10	Paid
BL000012	PT000831	AD00537	NULL	2025-07-18	14929.98	1422.24	343.37	1908.79	180.54	18784.92	Unpaid
BL000013	PT000936	AD02079	NULL	2025-08-29	2824.48	4215.96	1440.54	2838.69	231.30	20791.75	Partially P
BL000014	PT001492	AD02246	NULL	2025-03-25	9593.02	1296.19	3982.03	1967.47	88.79	23698.50	Partially P
BL000015	PT000383	AD02266	NULL	2025-07-21	7856.69	3871.98	3905.42	4238.38	973.71	20846.18	Unpaid
BL000019	PT001897	AD02176	NULL	2024-07-16	14224.63	1185.27	3867.74	1251.51	448.16	20977.31	Paid
BL000023	PT000381	AD01797	NULL	2024-10-05	9400.16	1833.85	3246.76	5593.42	1122.46	-21196.65	Paid
BL000024	PT001538	AD00829	NULL	2025-01-12	2170.73	3904.57	2219.50	2612.65	67.59	10975.04	Paid
BL000025	PT001516	AD01698	NULL	2024-01-27	3352.65	3790.27	1969.46	3688.71	67.97	18016.68	Paid
BL000029	PT000836	AD00457	NULL	2024-11-08	51.53	1894.46	726.30	1745.07	191.57	6452.50	Paid
BL000033	PT001557	AD00907	NULL	2024-03-09	11981.25	2686.13	1450.86	2120.96	106.29	18345.49	Partially P
BL000042	PT001340	AD01572	NULL	2025-04-27	1118.96	1482.38	3436.98	2749.66	251.34	9039.32	Paid
BL000043	PT000618	AD01216	NULL	2024-08-19	10814.15	2008.50	329.34	627.62	181.39	19545.40	Paid
BL000046	PT000573	AD01203	NULL	2025-05-23	12529.40	3429.10	264.72	4067.98	728.86	21020.06	Unpaid
BL000047	PT000868	AD01441	NULL	2025-10-06	13822.65	3476.08	3878.00	2519.31	896.66	34429.78	Paid
BL000048	PT000630	AD00462	NULL	2024-09-26	14967.55	1126.24	181.19	1090.07	1133.59	18498.64	Paid
BL000050	PT000590	AD01143	NULL	2025-03-28	11209.76	2859.61	577.05	5218.94	1136.32	21001.68	Unpaid
BL000051	PT000437	AD01195	NULL	2024-12-12	7545.54	3171.24	1451.81	1310.73	1057.33	17156.07	Partially P

Data cleaning:

Doctors data Cleaning

```
select gender,  
case  
when lower(TRIM(gender)) in ('male','m') then 'Male'  
when lower(TRIM(gender)) in ('female','f') then 'Female'  
else gender  
end as gender_clean  
from doctors;
```

```
3483      /* Doctors data Cleaning */  
3484  
3485 •   select gender,  
3486      case  
3487      when lower(TRIM(gender)) in ('male','m') then 'Male'  
3488      when lower(TRIM(gender)) in ('female','f') then 'Female'  
3489      else gender  
3490      end as gender_clean  
3491      from doctors;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	gender	gender_clean			
▶	Female	Female			
	Male	Male			
	Female	Female			
	Male	Male			
	Female	Female			
	Female	Female			
	Male	Male			
	Female	Female			
	Female	Female			
	Female	Female			
	Female	Female			
	Female	Female			
	Female	Female			
	Female	Female			
	Male	Male			
	Female	Female			
	Female	Female			

```
SET session SQL_SAFE_UPDATES = 0;
```

```
update doctors
set gender=
case
when lower(trim(gender)) in ('male','m') then 'Male'
when lower(TRIM(gender)) in ('female','f') then 'Female'
else gender
end;
```

```
SELECT
email AS old_email,
CASE
WHEN email LIKE '%gmail.com' AND email NOT LIKE '%@%'
THEN REPLACE(email,'gmail.com','@gmail.com')
WHEN email LIKE '%@gmail'
THEN REPLACE(email,'@gmail','@gmail.com') # THEN CONCAT(email,'.com')
WHEN email LIKE '%@@gmail.com'
THEN REPLACE(email,'@@gmail.com','@gmail.com')
ELSE email
END AS new_email
FROM Doctors;
```

```

3503 • SELECT
3504     email AS old_email,
3505     CASE
3506     WHEN email LIKE '%gmail.com' AND email NOT LIKE '%@%'
3507     THEN REPLACE(email,'gmail.com','@gmail.com')
3508     WHEN email LIKE '%@gmail'
3509     THEN REPLACE(email,'@gmail','@gmail.com') # THEN CONCAT(email,'.com')
3510     WHEN email LIKE '%@@gmail.com'
3511     THEN REPLACE(email,'@@gmail.com','@gmail.com')
3512     ELSE email
3513     END AS new_email
3514 FROM Doctors;
3515

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
old_email	new_email			
ramya.shetty2@outlook.com	ramya.shetty2@outlook.com			
shobha.panicker36@gmail.com	shobha.panicker36@gmail.com			
divya.achari89@yahoo.com	divya.achari89@yahoo.com			
devi.raman16@outlook.com	devi.raman16@outlook.com			
jayanthi.subramaniam82@yahoo.com	jayanthi.subramaniam82@yahoo.com			
latha.mudaliar32@gmail.com	latha.mudaliar32@gmail.com			
yogesh.balasubramaniam21@gmail.com	yogesh.balasubramaniam21@gmail.com			
indira.reddy19@outlook.com	indira.reddy19@outlook.com			
usha.pillai12@gmail.com	usha.pillai12@gmail.com			
ishwarya.reddy43@yahoo.com	ishwarya.reddy43@yahoo.com			
sandeep.nair98@rediffmail.com	sandeep.nair98@rediffmail.com			
bhavya.pillai16@rediffmail.com	bhavya.pillai16@rediffmail.com			

update Doctors

SET email=

CASE

WHEN email LIKE '%gmail.com' AND email NOT LIKE '%@%'

THEN REPLACE(email,'gmail.com','@gmail.com')

WHEN email LIKE '%@gmail'

THEN REPLACE(email,'@gmail','@gmail.com') # THEN CONCAT(email,'.com')

WHEN email LIKE '%@@gmail.com'

THEN REPLACE(email,'@@gmail.com','@gmail.com')

ELSE email

END

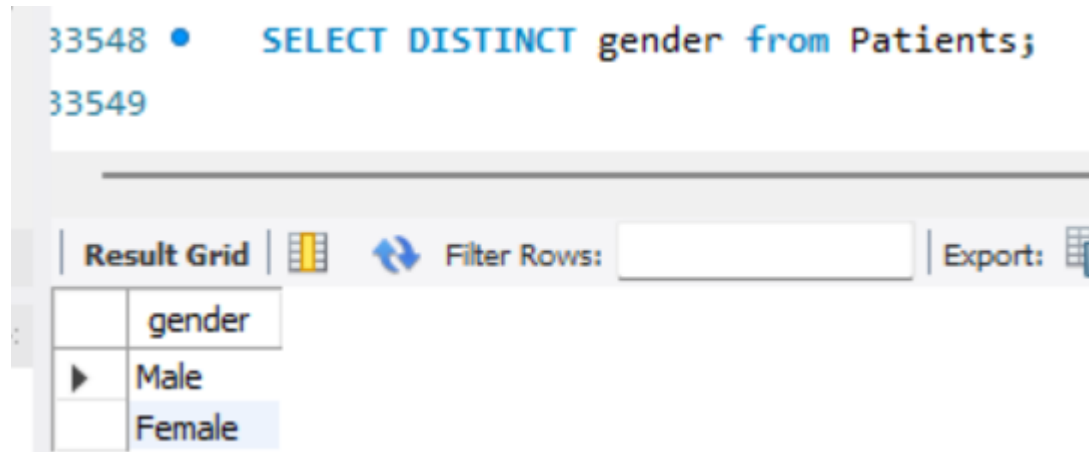
WHERE email not like '%@%.%' or

email like '%@@%.%';

Patients data Cleaning

```
UPDATE Patients  
SET first_name=ltrim(first_name)  
WHERE first_name <> LTRIM(first_name);
```

```
SELECT DISTINCT gender from Patients;
```



```
SELECT gender ,  
CASE  
WHEN lower(trim(gender)) in ('m','male')  
THEN 'Male'  
WHEN lower(trim(gender)) in ('f','female')  
THEN 'Female'  
END  
FROM Patients;
```

```

33550 • SELECT gender ,
33551 CASE
33552 WHEN lower(trim(gender)) in ('m','male')
33553 THEN 'Male'
33554 WHEN lower(trim(gender)) in ('f','female')
33555 THEN 'Female'
33556 END
33557 FROM Patients;

```

Result Grid			Filter Rows:	Export:	Wrap Cell
	gender	CASE WHEN lower(trim(gender)) in ('m','male') THEN 'Male' WHEN lower(trim(gender)) in ('f','female') THEN 'Female' END			
▶	Male	Male			
	Male	Male			
	Male	Male			
	Female	Female			
	Male	Male			
	Female	Female			
	Male	Male			

```

update Patients
set gender =
CASE
WHEN lower(trim(gender)) in ('m','male')
THEN 'Male'
WHEN lower(trim(gender)) in ('f','female')
THEN 'Female'
ELSE gender
END;

```

```

SELECT email AS old_email,
CASE
WHEN email LIKE '%gmail.com' AND email not like '%@%'
THEN REPLACE(email,'%gmail.com','%@gmail.com')
WHEN email like '%gmail'
then concat(email,'.com')
else email
end as new_email
FROM Patients;

```

```

}3570 •   SELECT email AS old_email,
}3571     CASE
}3572     WHEN email LIKE '%gmail.com' AND email not like '%@%'
}3573     THEN REPLACE(email,'%gmail.com','%@gmail.com')
}3574     WHEN email like '%gmail'
}3575     then concat(email,'.com')
}3576     else email
}3577     end as new_email
}3578     FROM Patients;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fel
	old_email	new_email				
▶	gopal.rao20@gmail.com	gopal.rao20@gmail.com				
	naveen.kamath56@gmail.com	naveen.kamath56@gmail.com				
	naveen.naicker4@yahoo.com	naveen.naicker4@yahoo.com				
	priya.raman91@outlook.com	priya.raman91@outlook.com				
	harish.venkatesh17@outlook.com	harish.venkatesh17@outlook.com				
	meenareddy@gmail.com	meenareddy@gmail.com				
	dinesh.natarajan12@yahoo.com	dinesh.natarajan12@yahoo.com				
	muthu.iyer9@yahoo.com	muthu.iyer9@yahoo.com				
	muthu.raman52@outlook.com	muthu.raman52@outlook.com				
	geetha.iyengar45@gmail.com	geetha.iyengar45@gmail.com				
	chandran.murthy88@rediffmail.com	chandran.murthy88@rediffmail.com				
	kavya.balasubramaniam76@gmai...	kavya.balasubramaniam76@gmai...				

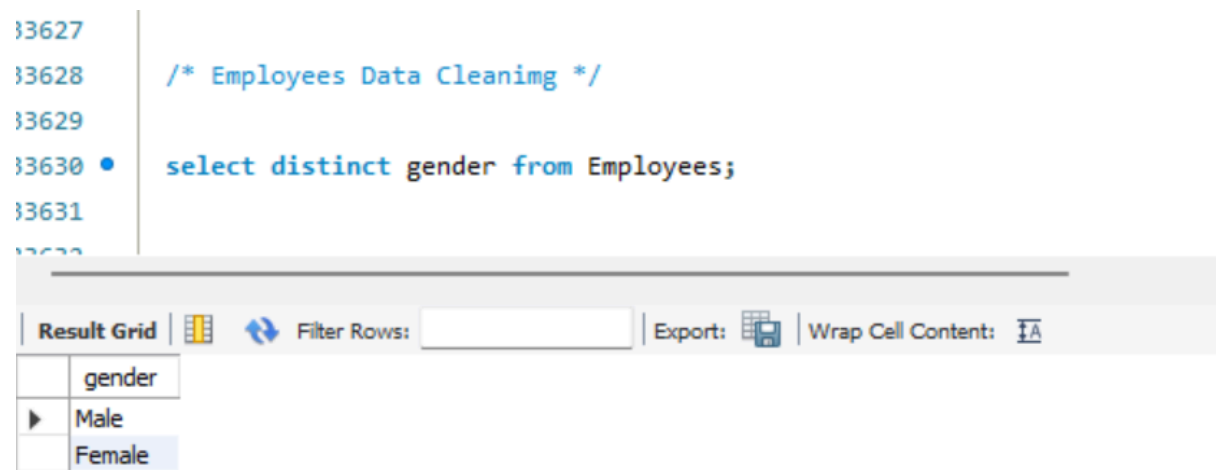
```

UPDATE Patients
set email=
CASE
WHEN email LIKE '%gmail.com' AND email not like '%@%'
THEN REPLACE(email,'gmail.com','@gmail.com')
WHEN email like '%gmail'
then concat(email,'.com')
else email
end
where email not like '%@%.%';

```

Employees

```
select distinct gender from Employees;
```



The screenshot shows a SQL IDE interface. The query editor on the left contains the following SQL code:

```

33627
33628      /* Employees Data Cleaning */
33629
33630 •   select distinct gender from Employees;
33631
33632

```

Below the query editor is a toolbar with the following options: "Result Grid" (selected), "Filter Rows:" (with a search box), "Export:" (with a download icon), and "Wrap Cell Content:" (with a text icon). Below the toolbar is a table with the following data:

	gender
▶	Male
	Female

```

UPDATE Employees
set gender =
CASE
when lower(trim(gender)) in ('male','m')
then 'Male'
when lower(trim(gender)) in ('female','f')
then 'Female'
else gender
end;

```

Business Analysis:

1.Which hospitals have the highest operational activity

```
SELECT h.hospital_id,h.hospital_name,COUNT(a.appointment_id) AS
total_appointments,COUNT(ad.admission_id) AS total_admissions
FROM hospitals h
LEFT JOIN appointments a
ON h.hospital_id = a.hospital_id
LEFT JOIN admissions ad
ON h.hospital_id = ad.hospital_id
GROUP BY h.hospital_id, h.hospital_name
order by total_appointments desc;
```

```
33822
33823  #Which hospitals have the highest operational activity
33824  • SELECT h.hospital_id,h.hospital_name,COUNT(a.appointment_id) AS total_appointments,COUNT(ad.admission_id) AS total_admissions
33825  FROM hospitals h
33826  LEFT JOIN appointments a
33827  ON h.hospital_id = a.hospital_id
33828  LEFT JOIN admissions ad
33829  ON h.hospital_id = ad.hospital_id
33830  GROUP BY h.hospital_id, h.hospital_name
33831  order by total_appointments desc;
```

hospital_id	hospital_name	total_appointments	total_admissions
H005	MediCare Belagavi North Campus	120292	120292
H004	MediCare Tiruchirapalli North Campus	16600	16600
H001	MediCare Kochi North Campus	15624	15624
H013	MediCare Chennai Institute of Medical Sciences	15015	15015
H007	MediCare Kozhikode North Campus	14784	14784
H014	MediCare Tirunelveli Central	14774	14774
H003	MediCare Vellore Annex	14276	14276
H029	MediCare Hubballi North Campus	14276	14276
H023	MediCare Erode Heights 2	14196	14196
H006	MediCare Madurai Superspeciality Centre	14193	14193
H021	MediCare Vellore Superspeciality Centre 4	14007	14007
H026	MediCare Thiruvananthapuram Superspeciality...	13851	13851
H027	MediCare Thiruvananthapuram Multispeciality ...	13448	13448
H008	MediCare Vellore Multispeciality Centre 2	13365	13365
H019	MediCare Vijayawada South Campus 2	13120	13120
H002	MediCare Coimbatore Central	13072	13072
H024	MediCare Thiruvananthapuram Annex 2	13050	13050
H009	MediCare Erode Annex	13040	13040
H010	MediCare Thrissur Heights	12782	12782
H016	MediCare Visakhapatnam Heights	12231	12231
H015	MediCare Guntur North Campus	12060	12060
H030	MediCare Thiruvananthapuram Institute of Me...	11856	11856

2. Which departments have the highest workload

```
SELECT d.department_id,h.hospital_name,d.department_name,COUNT(DISTINCT
a.appointment_id) AS total_appointments,COUNT(DISTINCT ad.admission_id) AS
total_admissions
FROM departments d
LEFT JOIN appointments a
ON d.hospital_id = a.hospital_id
LEFT JOIN admissions ad
ON d.department_id = ad.department_id
LEFT JOIN hospitals h
ON d.hospital_id = h.hospital_id
GROUP BY d.department_id,d.department_name;
```

33832

33833 #Which departments have the highest workload

33834 • SELECT d.department_id,h.hospital_name,d.department_name,COUNT(DISTINCT a.appointment_id) AS total_appointments,COUNT(DISTINCT ad.admiss:

33835 FROM departments d

33836 LEFT JOIN appointments a

33837 ON d.hospital_id = a.hospital_id

33838 LEFT JOIN admissions ad

33839 ON d.department_id = ad.department_id

33840 LEFT JOIN hospitals h

33841 ON d.hospital_id = h.hospital_id

33842 GROUP BY d.department_id,d.department_name;

33843

department_id	hospital_name	department_name	total_appointments	total_admissions
D001	MediCare Kochi North Campus	ENT	186	81
D002	MediCare Coimbatore Central	Emergency Medicine	172	75
D003	MediCare Vellore Annex	Ophthalmology	166	38
D004	MediCare Vellore Annex	Cardiology	166	45
D005	MediCare Tiruchirapalli North Campus	Neurology	166	96
D006	MediCare Belagavi North Campus	Ophthalmology	493	240
D007	MediCare Madurai Superspeciality Centre	Neurology	171	82
D008	MediCare Kozhikode North Campus	Urology	168	37
D009	MediCare Kozhikode North Campus	General Medicine	168	50
D010	MediCare Vellore Multispeciality Centre 2	Pulmonology	165	47
D011	MediCare Vellore Multispeciality Centre 2	Pediatrics	165	26
D012	MediCare Erode Annex	Pulmonology	163	78
D013	MediCare Thrissur Heights	General Surgery	154	80
D014	MediCare Salem South Campus	Oncology	165	57
D015	MediCare Vijayawada Central	Radiology	159	71
D016	MediCare Chennai Institute of Medical S...	Urology	165	52
D017	MediCare Chennai Institute of Medical S...	Dermatology	165	37

3. Which doctors have the highest workload

```
SELECT d.doctor_id,d.first_name, h.hospital_name,COUNT(a.appointment_id) AS
total_appointments,RANK() OVER (ORDER BY COUNT(a.appointment_id) DESC) AS
doctor_rank
FROM doctors d
LEFT JOIN appointments a
ON d.doctor_id = a.doctor_id
LEFT JOIN hospitals h
ON d.hospital_id = h.hospital_id
GROUP BY d.doctor_id,d.first_name;
```

```
33855
33856 • SELECT d.doctor_id,d.first_name, h.hospital_name,COUNT(a.appointment_id) AS total_appointments,RANK() OVER (ORDER BY COUNT(a.appointment_
33857 FROM doctors d
33858 LEFT JOIN appointments a
33859 ON d.doctor_id = a.doctor_id
33860 LEFT JOIN hospitals h
33861 ON d.hospital_id = h.hospital_id
33862 GROUP BY d.doctor_id,d.first_name;
33863
-----
```

doctor_id	first_name	hospital_name	total_appointments	doctor_rank
DR0008	Ramya	MediCare Tiruchirapalli North Campus	409	1
DR0268	Meena	MediCare Belagavi North Campus	52	2
DR0266	Yamuna	MediCare Belagavi North Campus	51	3
DR0177	Tarun	MediCare Belagavi North Campus	47	4
DR0408	Deepika	MediCare Belagavi North Campus	44	5
DR0277	Haritha	MediCare Belagavi North Campus	42	6
DR0151	Chandran	MediCare Belagavi North Campus	40	7
DR0259	Manasa	MediCare Belagavi North Campus	40	7
DR0325	Dinesh	MediCare Belagavi North Campus	40	7
DR0316	Yamuna	MediCare Belagavi North Campus	34	10
DR0361	Mohan	MediCare Belagavi North Campus	33	11
DR0228	Rajesh	MediCare Tirunelveli Central	32	12
DR0054	Vijay	MediCare Hubballi North Campus	30	13
DR0403	Sangeetha	MediCare Kochi North Campus	30	13
DR0230	Sandeep	MediCare Belagavi North Campus	29	15
DR0123	Yogesh	MediCare Kochi North Campus	27	16
DR0420	Vijay	MediCare Tirunelveli Central	27	16
DR0084	Bhavya	MediCare Kochi North Campus	25	18

Result 22 x

Read Only

4. Which patients have the highest healthcare activity

```
SELECT p.patient_id,p.first_name,h.hospital_name,h.city,COUNT(DISTINCT
a.appointment_id) AS total_appointments,COUNT(DISTINCT ad.admission_id) AS
total_admissions
FROM patients p
LEFT JOIN appointments a
ON p.patient_id = a.patient_id
LEFT JOIN admissions ad
ON p.patient_id = ad.patient_id
LEFT JOIN hospitals h
ON a.hospital_id = h.hospital_id
GROUP BY p.patient_id,p.first_name,h.hospital_name,h.city;
```

```
33863
33864 # Which patients have the highest healthcare activity
33865 • SELECT p.patient_id,p.first_name,h.hospital_name,h.city,COUNT(DISTINCT a.appointment_id) AS total_appointments,COUNT(DISTINCT ad.admission_id) AS total_admissions
33866 FROM patients p
33867 LEFT JOIN appointments a
33868 ON p.patient_id = a.patient_id
33869 LEFT JOIN admissions ad
33870 ON p.patient_id = ad.patient_id
33871 LEFT JOIN hospitals h
33872 ON a.hospital_id = h.hospital_id
33873 GROUP BY p.patient_id,p.first_name,h.hospital_name,h.city;
```

patient_id	first_name	hospital_name	city	total_appointments	total_admissions
PT000001	Gopal	MediCare Madurai Superspeciality Centre	Madurai	1	3
PT000001	Gopal	MediCare Thiruvananthapuram Multispeciality C...	Thiruvananthapuram	1	3
PT000002	Naveen	MediCare Tiruchirapalli North Campus	Tiruchirapalli	2	0
PT000003	Naveen	MediCare Hubballi North Campus	Hubballi	1	0
PT000004	Priya	MediCare Erode Annex	Erode	1	1
PT000005	Harish	MediCare Erode Heights 2	Erode	1	1
PT000006	Meena	MediCare Guntur North Campus	Guntur	1	2
PT000006	Meena	MediCare Warangal North Campus	Warangal	1	2
PT000007	Dinesh	MediCare Kozhikode North Campus	Kozhikode	1	1
PT000007	Dinesh	MediCare Madurai Central 2	Madurai	1	1
PT000007	Dinesh	MediCare Thiruvananthapuram Multispeciality C...	Thiruvananthapuram	1	1
PT000008	Muthu	MediCare Belagavi North Campus	Belagavi	1	0
PT000008	Muthu	MediCare Chennai Institute of Medical Sciences	Chennai	1	0
PT000008	Muthu	MediCare Kozhikode North Campus	Kozhikode	1	0
PT000008	Muthu	MediCare Tirunelveli Central	Tirunelveli	1	0
PT000009	Muthu	MediCare Erode Annex	Erode	1	2

5. Which hospitals have the highest number of admissions

```
SELECT h.hospital_name,COUNT(DISTINCT ad.admission_id) AS total_admissions
FROM hospitals h
JOIN admissions ad
ON h.hospital_id = ad.hospital_id
GROUP BY h.hospital_id, h.hospital_name
ORDER BY total_admissions DESC
LIMIT 5;
```

```
33874
33875 # Which hospitals have the highest number of admissions?
33876 • SELECT h.hospital_name,COUNT(DISTINCT ad.admission_id) AS total_admissions
33877 FROM hospitals h
33878 JOIN admissions ad
33879 ON h.hospital_id = ad.hospital_id
33880 GROUP BY h.hospital_id, h.hospital_name
33881 ORDER BY total_admissions DESC
33882 LIMIT 5;
```

Result Grid	Filter Rows:	Exports	Wrap Cell Content:	Fetch rows:
hospital_name		total_admissions		
▶	MediCare Belagavi North Campus	244		
	MediCare Tiruchirapalli North Campus	100		
	MediCare Erode Heights 2	91		
	MediCare Chennai Institute of Medical Sciences	91		
	MediCare Tirunelveli Central	89		

6. What are the admission patterns

```
SELECT admission_status,COUNT(*) AS total_admissions
FROM admissions
GROUP BY admission_status;
```

```
33883
33884 # What are the admission patterns?
33885 • SELECT admission_status,COUNT(*) AS total_admissions
33886 FROM admissions
33887 GROUP BY admission_status;
33888
```

admission_status	total_admissions
Admitted	371
Discharged	1869
Under Observation	260

7. Find the next admission for each patient

```
SELECT patient_id,admission_id,admission_date,LEAD(admission_date) OVER
(PARTITION BY patient_id ORDER BY admission_date) AS next_admission
FROM admissions
ORDER BY patient_id, admission_date;
```

```
33889
33890 # Find the next admission for each patient
33891 • SELECT patient_id,admission_id,admission_date,LEAD(admission_date) OVER (PARTITION BY patient_id ORDER BY admission_date) AS next_admiss:
33892 FROM admissions
33893 ORDER BY patient_id, admission_date;
```

patient_id	admission_id	admission_date	next_admission
PT000001	AD01080	2024-03-20	2024-03-23
PT000001	AD00763	2024-03-23	2024-05-31
PT000001	AD01110	2024-05-31	NULL
PT000004	AD02252	2024-04-08	NULL
PT000005	AD00747	2025-06-28	NULL
PT000006	AD02407	2024-09-08	2025-03-15
PT000006	AD01101	2025-03-15	NULL
PT000007	AD02454	2025-06-02	NULL
PT000009	AD00304	2024-02-21	2025-08-14
PT000009	AD01250	2025-08-14	NULL
PT000010	AD00341	2024-12-21	2025-06-01
PT000010	AD01738	2025-06-01	2025-07-02
PT000010	AD00405	2025-07-02	NULL
PT000011	AD00338	2024-07-29	2025-01-27
PT000011	AD01947	2025-01-27	2025-08-08
PT000011	AD02461	2025-08-08	NULL
PT000013	AD01886	2025-10-21	2025-11-10
PT000013	AD00779	2025-11-10	NULL
PT000014	AD01647	2025-03-25	2025-04-12
PT000014	AD00475	2025-04-12	2025-06-16
PT000014	AD00505	2025-06-16	NULL
PT000015	AD01209	2025-07-12	NULL
PT000016	AD00014	2024-08-18	2025-05-09

8. Are rooms being utilized effectively

```
SELECT h.hospital_id,h.hospital_name,COUNT(*) AS total_rooms,SUM(r.room_status =  
'Occupied') AS occupied_rooms,SUM(r.room_status = 'Vacant') AS available_rooms  
FROM rooms r  
LEFT JOIN hospitals h  
ON h.hospital_id = r.hospital_id  
GROUP BY hospital_id;
```

```
33894  
33895 # Are rooms being utilized effectively  
33896 • SELECT h.hospital_id,h.hospital_name,COUNT(*) AS total_rooms,SUM(r.room_status = 'Occupied') AS occupied_rooms,SUM(r.room_status = 'Vacant') AS available_rooms  
33897 FROM rooms r  
33898 LEFT JOIN hospitals h  
33899 ON h.hospital_id = r.hospital_id  
33900 GROUP BY hospital_id;  
33901
```

hospital_id	hospital_name	total_rooms	occupied_rooms	available_rooms
H001	MediCare Kochi North Campus	10	6	4
H002	MediCare Coimbatore Central	13	8	5
H003	MediCare Vellore Annex	40	25	15
H004	MediCare Tiruchirapalli North Campus	29	16	13
H005	MediCare Belagavi North Campus	47	27	20
H006	MediCare Madurai Superspeciality Centre	14	8	5
H007	MediCare Kozhikode North Campus	41	26	15
H008	MediCare Vellore Multispeciality Centre 2	11	7	4
H009	MediCare Erode Annex	29	17	11
H010	MediCare Thrissur Heights	39	22	15
H011	MediCare Salem South Campus	45	22	20
H012	MediCare Vijayawada Central	45	26	17
H013	MediCare Chennai Institute of Medical S...	35	15	20
H014	MediCare Tirunelveli Central	40	21	16
H015	MediCare Guntur North Campus	13	5	7
H016	MediCare Visakhapatnam Heights	20	12	7
H017	MediCare Vellore Superspeciality Centre 3	12	7	5
H018	MediCare Kozhikode Central 2	21	9	9
H019	MediCare Vijayawada South Campus 2	47	25	20
H020	MediCare Thiruvananthapuram Heights	9	5	4

9. Which treatments have the highest cost

```
SELECT treatment_name,COUNT(*) AS treatment_count,SUM(treatment_cost) AS  
total_cost  
FROM treatments  
GROUP BY treatment_name  
ORDER BY total_cost DESC  
LIMIT 10;
```

```
33901  
33902      # Which treatments have the highest cost  
33903 •    SELECT treatment_name,COUNT(*) AS treatment_count,SUM(treatment_cost) AS total_cost  
33904      FROM treatments  
33905      GROUP BY treatment_name  
33906      ORDER BY total_cost DESC  
33907      LIMIT 10;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
treatment_name	treatment_count	total_cost		
Appendicitis Management	213	6454746.17		
Type 2 Diabetes Mellitus Management	193	6011337.27		
Cataract Management	180	5925583.79		
Typhoid Fever Management	171	5831013.39		
Dengue Fever Management	193	5793849.76		
Stroke Management	189	5766452.33		
Fracture - Lower Limb Management	180	5715575.38		
Community Acquired Pneumonia Management	187	5513009.53		
Essential Hypertension Management	173	5489757.76		
Osteoarthritis Management	170	5489272.68		

10. Which laboratory tests have the highest count and cost

```
SELECT test_name,COUNT(*) AS test_count,SUM(test_cost) AS total_test_cost
FROM laboratory
GROUP BY test_name
order by total_test_cost desc;
```

33908

33909 # Which laboratory tests have the highest count and cost

33910 • SELECT test_name,COUNT(*) AS test_count,SUM(test_cost) AS total_test_cost

33911 FROM laboratory

33912 GROUP BY test_name

33913 order by total_test_cost desc;

33914

test_name	test_count	total_test_cost
▶ Ultrasound Abdomen	204	819076.38
HbA1c	190	805682.39
Liver Function Test	193	801558.47
Blood Culture	186	783125.36
COVID-19 RT-PCR	192	778872.13
ECG	175	769480.27
Vitamin D Test	177	743787.29
Blood Sugar PP	176	729340.27
Chest X-Ray	174	729008.04
Thyroid Profile	168	724365.53
MRI Brain	177	722392.83
Urine Routine	174	716738.64
Kidney Function Test	172	699918.42
Complete Blood Count	169	683385.26
CT Scan Abdomen	162	671015.73
Lipid Profile	166	661359.33
Blood Sugar Fasting	145	625399.59

11. Which medicines generate the highest pharmacy revenue

```
SELECT m.medicine_name,SUM(p.quantity) AS quantity_sold,SUM(p.total_price) AS  
total_revenue  
FROM medicines m  
JOIN pharmacy p  
ON m.medicine_id = p.medicine_id  
GROUP BY m.medicine_name  
ORDER BY total_revenue DESC;
```

```
33914  
33915 # Which medicines generate the highest pharmacy revenue  
33916 • SELECT m.medicine_name,SUM(p.quantity) AS quantity_sold,SUM(p.total_price) AS total_revenue  
33917 FROM medicines m  
33918 JOIN pharmacy p  
33919 ON m.medicine_id = p.medicine_id  
33920 GROUP BY m.medicine_name  
33921 ORDER BY total_revenue DESC;  
33922
```

Result Grid			
Filter Rows:			
Export:			
Wrap Cell Content:			
medicine_name	quantity_sold	total_revenue	
Atorvastatin 650mg	10003	4709198.63	
Vitamin B Complex 10mg	10573	4337437.89	
Paracetamol 10mg	8778	3953514.37	
Amoxicillin 40mg	9020	2245582.48	
Clopidogrel 500mg	9237	1405307.43	
Azithromycin 500mg	78	52006.66	
Vitamin D3 40mg	87	47251.59	
Insulin Glargine 10mg	56	47044.48	
Pantoprazole 250mg	72	46973.52	
Didofenac 10mg	55	43225.60	
Cetirizine 250mg	53	41129.42	
Enalapril 500mg	96	38137.38	
Warfarin 650mg	41	33763.78	
Paracetamol 500mg	50	33727.50	
Salbutamol 20mg	39	31671.51	
Vitamin D3 20mg	45	30964.35	
Enalapril 250mg	77	30130.04	
Iron Folic Acid 250mg	42	28642.88	
Atorvastatin 250mq	34	27852.80	

12. Bill status revenue

```
SELECT bill_status,COUNT(*) AS bill_count,SUM(total_amount) AS billed_amount
FROM billing
GROUP BY bill_status;
```

```
33926
33927 # Bill Status revenue
33928 • SELECT bill_status,COUNT(*) AS bill_count,SUM(total_amount) AS billed_amount
33929 FROM billing
33930 GROUP BY bill_status;
33931
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
bill_status	bill_count	billed_amount	
Paid	3267	45373660.99	
Unpaid	737	10024163.65	
Partially Paid	996	14277525.34	

13. Which payment methods perform best

```
SELECT payment_status,COUNT(payment_id) AS transaction_count,SUM(payment_amount) AS payment_amount
FROM payments
GROUP BY payment_status;
```

```
33931
33932 # Which payment methods perform best
33933 • SELECT payment_status,COUNT(payment_id) AS transaction_count,SUM(payment_amount) AS payment_amount
33934 FROM payments
33935 GROUP BY payment_status;
33936
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
payment_status	transaction_count	payment_amount	
Success	4413	61524505.02	
Refunded	197	2546467.93	
Failed	390	5500769.16	

14. Compare payment with previous payment

```
SELECT payment_id,patient_id,payment_date,payment_amount,LAG(payment_amount)
OVER(ORDER BY payment_date) AS previous_payment
FROM payments
ORDER BY payment_date;
```

33936
33937 # Compare payment with previous payment
33938 • SELECT payment_id,patient_id,payment_date,payment_amount,LAG(payment_amount) OVER(ORDER BY payment_date) AS previous_payment
33939 FROM payments
33940 ORDER BY payment_date;
33941

payment_id	patient_id	payment_date	payment_amount	previous_payment
PY003753	PT001885	2024-01-02	6344.38	6873.38
PY004547	PT001605	2024-01-03	4854.08	6344.38
PY004621	PT000396	2024-01-03	10569.49	4854.08
PY004359	PT000494	2024-01-03	16258.45	10569.49
PY004595	PT000089	2024-01-03	20982.13	16258.45
PY003855	PT000046	2024-01-03	6648.85	20982.13
PY001611	PT001841	2024-01-03	33153.91	6648.85
PY003394	PT000494	2024-01-03	16258.45	33153.91
PY002079	PT000046	2024-01-03	6648.85	16258.45
PY001716	PT000396	2024-01-03	10569.49	6648.85
PY003105	PT001797	2024-01-03	18472.29	10569.49
PY001549	PT000046	2024-01-03	6648.85	18472.29
PY002081	PT000971	2024-01-04	9272.69	6648.85
PY002680	PT000971	2024-01-04	9272.69	9272.69
PY003719	PT000410	2024-01-04	6528.08	9272.69
PY003791	PT000971	2024-01-04	9272.69	6528.08
PY004618	PT000938	2024-01-04	22932.57	9272.69
PY003787	PT000410	2024-01-04	6528.08	22932.57
PY004002	PT001479	2024-01-05	21460.41	6528.08
PY002413	PT001163	2024-01-05	5058.67	21460.41
PY000153	PT001224	2024-01-05	6046.27	5058.67
PY001975	PT001224	2024-01-05	6046.27	6046.27
PY000052	PT000991	2024-01-05	5161.32	6046.27

Result 48 x

KPI

1.Total Patients

```
SELECT COUNT(*) AS total_patients  
FROM patients;
```

```
33944  
33945      # Total Patients  
33946 •    SELECT COUNT(*) AS total_patients  
33947      FROM patients;  
33948
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	total_patients				
▶	2000				

2. Total Appointments

```
SELECT COUNT(appointment_id) AS total_appointments  
FROM appointments;
```

```
33948  
33949      # Total Appointments  
33950 •    SELECT COUNT(appointment_id) AS total_appointments  
33951      FROM appointments;  
33952
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	total_appointments				
▶	5100				

3. Total Admissions

```
SELECT COUNT(admission_id) AS total_admissions  
FROM admissions;
```

```
33952  
33953      # Total Admissions  
33954 •    SELECT COUNT(admission_id) AS total_admissions  
33955      FROM admissions;  
33956
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	total_appointments
▶	5100

4. Total Treatments

```
SELECT COUNT(treatment_id) AS total_treatments  
FROM treatments;
```

```
33956  
33957      # Total Treatments  
33958 •    SELECT COUNT(treatment_id) AS total_treatments  
33959      FROM treatments;  
33960
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	total_treatments
▶	3500

5. Total Treatment Cost

```
SELECT SUM(treatment_cost) AS total_treatment_cost
FROM treatments;
```

```
33960
33961      # Total Treatment Cost
33962 •    SELECT SUM(treatment_cost) AS total_treatment_cost
33963      FROM treatments;
33964
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_treatment_cost			
▶	107899741.54			

6. Total Lab Activity

```
SELECT COUNT(lab_test_id) AS total_lab_activity
FROM laboratory;
```

```
33964
33965      # Total Lab Activity
33966 •    SELECT COUNT(lab_test_id) AS total_lab_activity
33967      FROM laboratory;
33968
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_lab_activity			
▶	3000			

7. Total Lab Cost

```
SELECT SUM(test_cost) AS total_lab_cost
FROM laboratory;
```

```
33968
33969      # Total Lab Cost
33970 •    SELECT SUM(test_cost) AS total_lab_cost
33971      FROM laboratory;
33972
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	total_lab_cost				
▶	12464505.93				

8. Total Pharmacy Activity

```
SELECT COUNT(pharmacy_sale_id) AS total_pharmacy_activity
FROM pharmacy;
```

```
33972
33973      # Total Pharmacy Activity
33974 •    SELECT COUNT(pharmacy_sale_id) AS total_pharmacy_activity
33975      FROM pharmacy;
33976
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	total_lab_cost				
▶	12464505.93				

9. Pharmacy Revenue

```
SELECT SUM(total_price) AS pharmacy_revenue  
FROM pharmacy;
```

```
33976  
33977      # Pharmacy Revenue  
33978 •    SELECT SUM(total_price) AS pharmacy_revenue  
33979      FROM pharmacy;  
33980
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	total_lab_cost				
▶	12464505.93				

10. Total Billed Amount

```
SELECT SUM(total_amount) AS total_billed_amount  
FROM billing;
```

```
33980  
33981      # Total Billed Amount  
33982 •    SELECT SUM(total_amount) AS total_billed_amount  
33983      FROM billing;  
33984
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	total_lab_cost				
▶	12464505.93				

11. Total Payment Collected

```
SELECT SUM(payment_amount) AS total_payment_collected  
FROM payments;
```

33984

33985 # Total Payment Collected

33986 • SELECT SUM(payment_amount) AS total_payment_collected

33987 FROM payments;

33988

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	total_payment_collected
▶	69571742.11